Indian Institute Of Information Technology Guwahati

Project Report 2

Group Number - 14

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Relational Design Report

Introduction:
This document presents the relational schema derived from the ER model of the Real Estate Management System. The schema has been refined to eliminate redundancy, enforce integrity, and ensure normalization up to higher normal forms.

Relations (Tables Definition)
(Primary Keys are underlined, Foreign Keys are specified clearly)

1. *User(**UID**, Name, Email, Phone, Password, ProfileURL, RegDate, Area, City, Pin)*
   > Stores all platform users including buyers, sellers, and agents (agents are specialized users).

2. *Agent(**AID**, OfficeID, Rate, UserRating, License, Status)*
   > FK → AID referencing User(UID)
   > Stores additional details specific to agents.

3. *Office(**OfficeID**, Password, Email, Telephone, City)*
   > Represents real estate offices managing agents and listings.

4. *Property(**PID**, OwnerID, OfficeID, HouseNo, Locality, Area, City, Pin, Size, Type, BHK, YearBuilt)*
   > FK → OwnerID referencing User(UID), OfficeID
   > Stores details of properties (houses/apartments)

5. ListingToken(**TokenID**, PID, AID, ListingDate, ListType, Price, Description, Status)
   > FK → PID, AID
   > Represents listings (rent/sell) created by agents for properties.

6. *Transaction(**TID**, AID, TokenID, BuyerID, Amount, Date, Type, Mode)*
   > FK → AID, TokenID, BuyerID referencing User(UID)
   > Stores completed deals (rent/sale transactions).

7. P*erformance(**TrackID**, AID, Deals, TotalSales, Score, UserRating, DealsLeft)*
   > FK → AID
   > Tracks agent performance metrics.

8. *Notification(**CODE**, SID, RID, Message, Type, Date, Time, Status)*
   > FK → SID, RID referencing User(UID)
   > Stores notifications/messages between users.

9. *Images(**ImgID**, PID, URL, Date)*
   > FK → *PID*
   > Stores property images.

Normalization

Functional Dependencies (FDs)

Examples:

- UID → Name, Email, Phone, Password, ProfileURL, RegDate, Area, City, Pin
- AID → OfficeID, CommissionRate, LicenseNumber, Experience, Status
- PID → OwnerID, OfficeID, HouseNo, Locality, Area, City, Pin, Size, Type, BHK, YearBuilt
- TrackID → Deals, Score, TotalSales, UserRating, DealsLeft
- OfficeID → Email, Telephone, City
- TokenID → PID, AID, ListingDate, ListType, Price, Description, Status
- TID → AID, TokenID, BuyerID, Amount, Date, Type, Mode

First Normal Form (1NF):

A relation is in 1NF if,

- All attributes contain atomic values
- No repeating groups exist
- Each record has a primary key

Evaluation: All relations (USER, AGENT, PROPERTY, LISTINGTOKEN, TRANSACTION, PERFORMANCE, NOTIFICATION, IMAGES, OFFICE)
contain atomic attributes and defined primary keys. Therefore all tables satisfy 1NF.

Second Normal Form (2NF):

A relation is in 2NF if,

- It is already in 1NF
- No partial dependency.
  All non-key attributes are fully dependent on the entire primary key

Evaluation: All tables use single attribute primary keys (UID, AID, PID, TokenID, TID, TrackID, Code, ImgID, OfficeID).
Since there are no composite keys, partial dependencies cannot occur.
Thus all relations satisfy 2NF.

Third Normal Form (3NF):

A relation is in 3NF if,

- It is in 2NF
- No transitive dependency exists
  (non-key attribute determining another non-key attribute).

Evaluation: All the non-key attributes depend on the primary key (UID, AID, PID, TokenID, TID, TrackID, Code, ImgID, OfficeID) and not on any other non-key attribute. Thus it holds the 3NF.

Boyce-Codd Normal Form (BCNF)

- A relation is in BCNF if for every functional dependency X → Y, X is a superkey of the relation.
- If LHS is not the superkey then it is not in BCNF

Evaluation:
Since all the functional dependency are of type: PK → all other attributes, where PK is the Primary Key, Hence the LHS will always be Super Key thus all relations satisfies BCNF.

Observations and Improvements

Possible improvements:

- Removed some derived attributes from the previous model
- Optional address normalization by introducing an ADDRESS relation.
- Optional geographic normalization by introducing a PINCODE relation if strict city-pin

Individual Contribution

1. Anmol Kumar (2401031)

   Checked the data consistency and matched the constraint of the data, suggested possible edits

2. Anmol Kumar (2401032)

   Updated the tables with the populated data of the spreadsheet, and designed the backend

3. Ansik Singh Tomar (2401037)

   Prepared a spreadsheet with all the populated data, keeping the constraints in mind and drafted this report

4. Anuved Pratap Singh (2401041)

   Written the java code and mapped the Entities using JPA to form the database Schema for the mysql

Schema

Performance	
PK	TrackID
FK	AID
	Deals
	TotalSales
	Score
	UserRating
	DealsLeft

Notification	
PK	CODE
FK	SID
FK	RID
	Message
	Type
	Date
	Time
	Status (read/unread)

ListingToken	
PK	TokenID
FK	PID
FK	AID
	Listing Date
	ListType (Rent/Sell)
	Price
	Description
	Status (Open/Closed)

Agent	
PK - FK	AID (=UID)
FK	OfficeID
	Rate
	Status (Active/Inactive)

Property	
PK	PID
FK	OwnerID
FK	OfficeID
	HouseNo
	Locality
	Area
	City
	Pin
	Size
	Type (Flat/ Appartment)
	BHK
	Year Built

Images	
PK	ImgID
FK	PID
	URL
	Date

Office	
PK	OfficeID
	Password
	Email
	TelePhone
	City

Transaction	
PK	TID
FK	AID
FK	TokenID
FK	BuyerID
	Amount
	Date
	Type
	Mode

USER	
PK	UID
	Name
	Email
	Phone
	Password
	ProfileURL
	RegDate
	Area
	City
	Pin